

tfidf-baselines

tfidf-baselines 20251209-1400

```
LogReg metrics: ({'subset_accuracy': 0.8136482939632546, 'micro_f1': 0.8967836871978138,
'macro_f1': 0.84444800057691262, 'hamming_loss': 0.025774278215223097}),      label      f1
0   2436   0.949367
1   2434   0.940919
2   2441   0.930748
3   2442   0.919003
4   2439   0.854460
5   2445   0.845771
6   2447   0.830189
7   2438   0.735178
8   2444   0.734940
9   2451   0.704225)
LinearSVC metrics: ({'subset_accuracy': 0.8792650918635171, 'micro_f1':
0.9277703005832212, 'macro_f1': 0.8926912835783638, 'hamming_loss': 0.016902887139107613},
      label      f1
0   2441   0.966715
1   2436   0.960526
2   2434   0.956650
3   2442   0.932794
4   2451   0.877193
5   2447   0.868613
6   2439   0.861702
7   2445   0.844444
8   2438   0.839378
9   2444   0.818898
```

su_admin_BERT_dist_pred

su_admin_BERT_dist_pred 2025-12-08 21:20

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su_admin_BERT_dist_pred 2025-12-08 21:20
=====
FINAL EVALUATION METRICS
=====

eval_loss: 0.2162
eval_mae_pct: 1.3943
eval_top1_accuracy: 0.9286
eval_mean_cosine_sim: 0.9509
eval_mean_js_divergence: 0.0774
eval_runtime: 34.9933
eval_samples_per_second: 54.4390
eval_steps_per_second: 1.7150
epoch: 3.0000
```

su_admin_BERT_dist_pred 2025-12-08 23:00

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FINAL EVALUATION METRICS

=====

eval_loss: 0.2158
eval_mae_pct: 1.4367
eval_top1_accuracy: 0.9239
eval_mean_cosine_sim: 0.9511
eval_mean_js_divergence: 0.0794
eval_runtime: 35.3052
eval_samples_per_second: 53.9580
eval_steps_per_second: 1.6990
epoch: 3.0000

Per-Label MAE (percentage points):

uo_code	uo_name	mae_pct
2442	SA	4.222667
2434	HU	2.198112
2441	NA	2.152678
2438	LU	1.809316
2447	ÖV	1.460524
2439	ME	0.742094
2445	VÅ	0.668917
2444	TE	0.606430
2436	JU	0.328126
2451	VU	0.178244

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METHOD COMPARISON SUMMARY

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Compare these results with:

1. Binary BERT (post-hoc normalized):
 - MAE: 2.75, Top-1: 85.1%, Cosine: 0.94
2. Direct Distributional BERT (this model):
 - MAE: 1.44, Top-1: 92.4%, Cosine: 0.9511

lis070-su-admin-BERT-binary 20251210 18:30

```
{'eval_loss': 0.05396690219640732,  
'eval_subset_accuracy': 0.9076115485564304,  
'eval_micro_f1': 0.9354697102721685,  
'eval_macro_f1': 0.9046030552413425,  
'eval_hamming_loss': 0.015433070866141733,  
'eval_runtime': 34.2307,  
'eval_samples_per_second': 55.652,  
'eval_steps_per_second': 1.753,  
'epoch': 3.0}
```

```
=====
DISTRIBUTIONAL EVALUATION METRICS
=====
Mean Absolute Error (percentage points): 2.79
Mean Cosine Similarity:                  0.9382
Mean Jensen-Shannon Divergence:         0.1915
Top-1 Accuracy (primary label match):    0.8525
```

Per-Label MAE (percentage points):

uo_code	mae_pct
2442	7.022961
2434	3.773409
2441	3.452049
2438	2.910137
2447	2.828696
2444	2.301982
2439	1.794413
2445	1.479483
2436	1.377615
2451	0.960420

```
=====
WORST PREDICTIONS (highest MAE)
=====
```

Course ID: 31197

Gold:	{2447: 100.0}
Gold dist:	[0. 0. 0. 0. 0. 0. 0. 0. 100. 0.]
Pred dist:	[96.5 0.3 0.4 0.3 0.6 0.4 0.3 0.3 0.5 0.3]
MAE:	19.91

Course ID: 14747

Gold:	{2438: 100.0}
Gold dist:	[0. 0. 100. 0. 0. 0. 0. 0. 0. 0.]
Pred dist:	[96.8 0.3 0.5 0.2 0.4 0.7 0.3 0.2 0.4 0.3]
MAE:	19.90

Course ID: 14747

Gold:	{2438: 100.0}
Gold dist:	[0. 0. 100. 0. 0. 0. 0. 0. 0. 0.]
Pred dist:	[96.8 0.3 0.5 0.2 0.5 0.6 0.3 0.2 0.4 0.3]
MAE:	19.90

Course ID: 49441

Gold:	{2436: 100.0}
Gold dist:	[0. 100. 0. 0. 0. 0. 0. 0. 0. 0.]
Pred dist:	[25.9 0.6 0.4 0.3 0.2 71.4 0.2 0.3 0.4 0.2]
MAE:	19.88

Course ID: 43983

Gold:	{2434: 100.0}
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```
Gold dist: [100.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
Pred dist: [ 0.6  0.5  0.4 45.6  0.6 49.8  1.1  0.4  0.5  0.5]
MAE:      19.87
```

Course ID: 43983

```
Gold:      {2434: 100.0}
Gold dist: [100.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
Pred dist: [ 0.6  0.5  0.4 45.6  0.6 49.8  1.1  0.4  0.5  0.5]
MAE:      19.87
```

Course ID: 26238

```
Gold:      {2441: 100.0}
Gold dist: [  0.  0.  0.  0. 100.  0.  0.  0.  0.  0.]
Pred dist: [59.2  0.4 29.6  0.4  0.9  1.4  0.3  0.4  6.7  0.8]
MAE:      19.82
```

Course ID: 15798

```
Gold:      {2447: 100.0}
Gold dist: [  0.  0.  0.  0.  0.  0.  0.  0. 100.  0.]
Pred dist: [96.4  0.2  0.4  0.2  0.7  0.5  0.2  0.2  0.9  0.3]
MAE:      19.82
```

Course ID: 15798

```
Gold:      {2447: 100.0}
Gold dist: [  0.  0.  0.  0.  0.  0.  0.  0. 100.  0.]
Pred dist: [96.4  0.2  0.4  0.2  0.7  0.5  0.2  0.2  0.9  0.3]
MAE:      19.82
```

Course ID: 26238

```
Gold:      {2441: 100.0}
Gold dist: [  0.  0.  0.  0. 100.  0.  0.  0.  0.  0.]
Pred dist: [60.6  0.4 29.1  0.3  1.  1.3  0.3  0.4  5.8  0.9]
MAE:      19.80
```

Per-Label MAE with names:

uo_code	uo_name	mae_pct	
2442	SA	7.022961	# Samhällsvetenskap
2434	HU	3.773409	# Humaniora
2441	NA	3.452049	# Naturvetenskap
2438	LU	2.910137	# Lärarutbildning
2447	ÖV	2.828696	# Övrigt
2444	TE	2.301982	# Teknik
2439	ME	1.794413	# Medicin
2445	VÅ	1.479483	# Vård
2436	JU	1.377615	# Juridik
2451	VU	0.960420	# Verksamhetsförlagd utbildning

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LABEL CONFUSION ANALYSIS

(When gold is X, model predicts Y instead)

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gold_code	gold_name	pred_code	pred_name	confusion_pct	n_cases
2444	TE	2442	SA	94.594595	70
2445	VÅ	2442	SA	85.185185	23
2439	ME	2442	SA	84.507042	60
2445	VÅ	2438	LU	14.814815	4
2438	LU	2442	SA	12.328767	9

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OVERALL SUMMARY

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Primary label distribution (Gold vs Predicted):

HU	: Gold= 677, Pred= 671, Diff= -6
JU	: Gold= 77, Pred= 76, Diff= -1
LU	: Gold= 73, Pred= 87, Diff= +14
ME	: Gold= 71, Pred= 10, Diff= -61
NA	: Gold= 321, Pred= 316, Diff= -5
SA	: Gold= 439, Pred= 588, Diff=+149
TE	: Gold= 74, Pred= 0, Diff= -74
VÅ	: Gold= 27, Pred= 0, Diff= -27
ÖV	: Gold= 123, Pred= 135, Diff= +12
VU	: Gold= 23, Pred= 22, Diff= -1

Per-label primary accuracy:

HU	: 96.5% (677 cases)
JU	: 93.5% (77 cases)
LU	: 82.2% (73 cases)
ME	: 14.1% (71 cases)
NA	: 91.9% (321 cases)
SA	: 90.9% (439 cases)
TE	: 0.0% (74 cases)
VÅ	: 0.0% (27 cases)
ÖV	: 91.9% (123 cases)
VU	: 95.7% (23 cases)

Bootstrap 2025-12-12

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COMPARISON 1: Linear SVC vs KB-BERT (Multi-label Classification)

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Metric: Subset Accuracy (↑ = better)

=====

```
Linear SVC          : 82.94% (95% CI: [81.21, 84.62])
KB-BERT             : 90.76% (95% CI: [89.45, 92.02])
-----

Difference (B - A): +7.82% (95% CI: [+6.30, +9.34])
p-value: 0.0002 *
Significant at  $\alpha=0.05$ : Yes
Winner: KB-BERT

=====

Metric: Subset Accuracy ( $\uparrow$  = better)
=====

Linear SVC          : 82.94% (95% CI: [81.21, 84.62])
KB-BERT             : 90.76% (95% CI: [89.45, 92.02])
-----

Difference (B - A): +7.82% (95% CI: [+6.30, +9.34])
p-value: 0.0002 *
Significant at  $\alpha=0.05$ : Yes
Winner: KB-BERT

=====

Metric: Micro-F1 ( $\uparrow$  = better)
=====

Linear SVC          : 89.97% (95% CI: [88.87, 91.02])
KB-BERT             : 93.55% (95% CI: [92.52, 94.52])
-----

Difference (B - A): +3.57% (95% CI: [+2.56, +4.58])
p-value: 0.0002 *
Significant at  $\alpha=0.05$ : Yes
Winner: KB-BERT

=====

Metric: Micro-F1 ( $\uparrow$  = better)
=====

Linear SVC          : 89.97% (95% CI: [88.87, 91.02])
KB-BERT             : 93.55% (95% CI: [92.52, 94.52])
-----

Difference (B - A): +3.57% (95% CI: [+2.56, +4.58])
p-value: 0.0002 *
Significant at  $\alpha=0.05$ : Yes
Winner: KB-BERT

=====

Metric: Macro-F1 ( $\uparrow$  = better)
=====

Linear SVC          : 85.77% (95% CI: [83.59, 87.59])
KB-BERT             : 90.46% (95% CI: [88.49, 92.12])
-----

Difference (B - A): +4.69% (95% CI: [+3.27, +6.23])
p-value: 0.0002 *
Significant at  $\alpha=0.05$ : Yes
Winner: KB-BERT

=====
```

Metric: Macro-F1 ($\uparrow$ = better)

```
=====
Linear SVC          : 85.77% (95% CI: [83.59, 87.59])
KB-BERT             : 90.46% (95% CI: [88.49, 92.12])
-----
```

Difference (B - A): +4.69% (95% CI: [+3.27, +6.23])

p-value: 0.0002 *

Significant at $\alpha=0.05$: Yes

Winner: KB-BERT

Metric: Hamming Loss ($\downarrow$ = better)

```
=====
Linear SVC          : 2.31% (95% CI: [2.07, 2.57])
KB-BERT             : 1.54% (95% CI: [1.31, 1.79])
-----
```

Difference (B - A): -0.77% (95% CI: [-1.00, -0.54])

p-value: 0.0002 *

Significant at $\alpha=0.05$: Yes

Winner: KB-BERT

Metric: Hamming Loss ($\downarrow$ = better)

```
=====
Linear SVC          : 2.31% (95% CI: [2.07, 2.57])
KB-BERT             : 1.54% (95% CI: [1.31, 1.79])
-----
```

Difference (B - A): -0.77% (95% CI: [-1.00, -0.54])

p-value: 0.0002 *

Significant at $\alpha=0.05$: Yes

Winner: KB-BERT

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COMPARISON 2: BERT Binary vs BERT Distributional (Distribution Prediction)

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Note: To run this comparison, you need BERT binary sigmoid probabilities.
These should be saved during BERT binary model inference:

```
Y_prob = 1 / (1 + np.exp(-logits)) # sigmoid
np.save('bert_binary_probs.npy', Y_prob)
```

Metric: Top-1 Accuracy ($\uparrow$ = better)

```
=====
BERT Binary          : 85.25% (95% CI: [83.62, 86.77])
BERT Distributional  : 92.39% (95% CI: [91.18, 93.54])
-----
```

Difference (B - A): +7.14% (95% CI: [+5.83, +8.56])

p-value: 0.0002 *

Significant at $\alpha=0.05$: Yes

Winner: BERT Distributional

```

=====
Metric: Top-1 Accuracy (↑ = better)
=====
BERT Binary          : 85.25% (95% CI: [83.62, 86.77])
BERT Distributional  : 92.39% (95% CI: [91.18, 93.54])
-----
Difference (B - A): +7.14% (95% CI: [+5.83, +8.56])
p-value: 0.0002 *
Significant at α=0.05: Yes
Winner: BERT Distributional

=====
Metric: MAE (percentage points) (↓ = better)
=====
BERT Binary          : 2.7901 (95% CI: [2.6109, 2.9757])
BERT Distributional  : 1.4367 (95% CI: [1.2597, 1.6191])
-----
Difference (B - A): -1.3534 (95% CI: [-1.4668, -1.2402])
p-value: 0.0002 *
Significant at α=0.05: Yes
Winner: BERT Distributional

=====
Metric: MAE (percentage points) (↓ = better)
=====
BERT Binary          : 2.7901 (95% CI: [2.6109, 2.9757])
BERT Distributional  : 1.4367 (95% CI: [1.2597, 1.6191])
-----
Difference (B - A): -1.3534 (95% CI: [-1.4668, -1.2402])
p-value: 0.0002 *
Significant at α=0.05: Yes
Winner: BERT Distributional

=====
Metric: Mean Cosine Similarity (↑ = better)
=====
BERT Binary          : 0.9382 (95% CI: [0.9294, 0.9466])
BERT Distributional  : 0.9511 (95% CI: [0.9427, 0.9590])
-----
Difference (B - A): +0.0129 (95% CI: [+0.0079, +0.0180])
p-value: 0.0002 *
Significant at α=0.05: Yes
Winner: BERT Distributional

=====
Metric: Mean Cosine Similarity (↑ = better)
=====
BERT Binary          : 0.9382 (95% CI: [0.9294, 0.9466])
BERT Distributional  : 0.9511 (95% CI: [0.9427, 0.9590])
-----
Difference (B - A): +0.0129 (95% CI: [+0.0079, +0.0180])
p-value: 0.0002 *
Significant at α=0.05: Yes

```


Winner: BERT Distributional

=====									
INDIVIDUAL MODEL CONFIDENCE INTERVALS									
=====									
Model		Subset Accuracy		Micro-F1		Macro-F1			
Hamming Loss									
Logistic Regression		77.80%	[75.91, 79.69]	87.77%	[86.64, 88.86]	83.64%	[81.52, 85.49]		
		3.02%	[2.74, 3.31]						
Linear SVC		82.94%	[81.21, 84.62]	89.97%	[88.87, 91.02]	85.77%	[83.59, 87.59]		
		2.31%	[2.07, 2.57]						
KB-BERT		90.76%	[89.45, 92.02]	93.55%	[92.52, 94.52]	90.46%	[88.49, 92.12]		
		1.54%	[1.31, 1.79]						
Model		Subset Accuracy		Micro-F1		Macro-F1			
Hamming Loss									
Logistic Regression		77.80%	[75.91, 79.69]	87.77%	[86.64, 88.86]	83.64%	[81.52, 85.49]		
		3.02%	[2.74, 3.31]						
Linear SVC		82.94%	[81.21, 84.62]	89.97%	[88.87, 91.02]	85.77%	[83.59, 87.59]		
		2.31%	[2.07, 2.57]						
KB-BERT		90.76%	[89.45, 92.02]	93.55%	[92.52, 94.52]	90.46%	[88.49, 92.12]		
		1.54%	[1.31, 1.79]						